Project Design Phase

Problem - Solution Fit Template

Date: 01 October 2026

Team ID: SWTID-2026-8643

Project Name: Script-Controlled ACL - Restrict Record Access Based on Field Value

Maximum Marks: 2 Marks

The Problem-Solution Fit shows that the identified customer problem is clearly understood and that the proposed solution addresses it. It helps the team identify customer behaviour, needs and the reasons a solution is likely to work.

Purpose:

• Solve complex problems in a way that fits the customer's situation.

• Improve solution adoption by working with existing behaviour, channels and practices.

• Strengthen communication and messaging around the customer's needs.

• Build trust by solving frequent, urgent or costly problems.

• Understand the current situation so it can be improved for the target group.

1. CUSTOMER SEGMENT(S) [CS]

• ServiceNow administrators and developers managing institutional or student data

• Branch staff and faculty (for example, EEE) who need access only to their branch records

• Institution administrators responsible for student-data privacy

2. JOBS-TO-BE-DONE / PROBLEMS [J&P]

• Control who can view, create, edit and delete records

• Ensure EEE users see only EEE records while administrators retain full access

• Protect sensitive student information such as email, phone and faculty details

3. TRIGGERS [TR]

• A user notices student records belonging to another branch

• A data-privacy or audit requirement is raised

• A new table containing sensitive information needs protection

4. EMOTIONS: BEFORE / AFTER [EM]

• Before: worried about data exposure and uncertain about who can view or change records

• After: more confident, secure and in control, with each role receiving the access it needs

5. AVAILABLE SOLUTIONS [AS]

• Basic role-based ACLs: provide table-level control but not field-value filtering

• Manual access verification

• Separate tables or instances for each branch: expensive and difficult to maintain

• Pros: simple. Cons: inflexible, less scalable and without proper record-level control

6. CUSTOMER CONSTRAINTS [CC]

• security_admin access is required to create or modify ACLs

• Advanced ACLs require scripting knowledge

• The configuration must not interrupt access for administrators or legitimate users

• Limited time and no budget for additional tools

7. BEHAVIOUR [BE]

• Administrators create custom roles and assign them to users

• Access is tested by impersonating different users

• List views are checked to confirm what each user can see

• ACL conditions are updated when access requirements change

8.1 ONLINE (CHANNELS OF BEHAVIOUR) [CH]

• ServiceNow instance: Users, Roles, Tables and Access Control (ACL) modules

• ServiceNow documentation, community forums and online tutorials

8.2 OFFLINE [CH]

• Team and department-head discussions about who should access specific records

• Written access policies and periodic review meetings

9. PROBLEM ROOT CAUSE [RC]

• Table-level access does not separate records according to the Branch field

• Without record-level rules, users with table access can see records from all branches

• Read, create, write and delete permissions are not always configured independently

10. YOUR SOLUTION [SL]

• Script-controlled ACLs on the u_institution_details table

• Read ACL (bb1): Branch = EEE condition plus an advanced script; administrators receive full access, bb1 users see EEE records,and other users are denied

• Create (bb2), Write (bb3) and Delete (bb4) ACLs, each controlled by its own role

• Validation through impersonation of the EEE user, a user without roles and an administrator• Result: record-level security without duplicating tables

References:

1. IdeaHacker Network – Problem-Solution Fit Canvas

2. Medium – Problem-Solution Fit Canvas